

Project Design Phase

Project Title: Diabetic Retinopathy Detection Using Deep Learning

1. Introduction

The project design phase focuses on defining the structure and architecture of the proposed system. In this phase, the system components, data flow, and working process of the application are planned. Proper system design ensures that the project is implemented efficiently and meets the required objectives.

The main goal of this project is to design a system that can automatically detect diabetic retinopathy from retinal images using deep learning techniques.

2. System Architecture

The proposed system consists of the following major components:

Input Module

The user uploads retinal images into the system for analysis.

Image Preprocessing Module

The input images are resized and normalized to improve image quality and prepare them for processing by the deep learning model.

Feature Extraction Module

Important features from retinal images are extracted automatically using a Convolutional Neural Network (CNN).

Classification Module

The deep learning model analyzes the extracted features and classifies the image as either diabetic retinopathy or normal.

Output Module

The system displays the prediction result to the user.

3. System Workflow

- The working process of the system follows these steps:
- User uploads a retinal image.
- The system preprocesses the image.
- The deep learning model extracts features from the image.
- The model classifies the image.
- The result is displayed to the user.

4. Technologies Used

- The system uses the following technologies:
- Programming Language: Python
- Deep Learning Framework: TensorFlow / Keras
- Image Processing Library: OpenCV
- Development Environment: Jupyter Notebook
- Version Control: GitHub

5. Advantages of the System

- Faster disease detection
- High accuracy using deep learning algorithms
- Automated analysis of medical images
- Supports early diagnosis and treatment

5. Conclusion

The project design phase defines the architecture and workflow of the proposed system. By using deep learning techniques, the system can effectively analyze retinal images and detect diabetic retinopathy, assisting doctors in early diagnosis and improving healthcare services.